

Calandra Tier 1 Data Platform Application

PPUK Data Platform

Technical Requirements & Deployment Best Practices

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Table of Contents

<u>1.</u>	<u>OVERVIEW</u>	<u>4</u>
<u>2.</u>	<u>HIGH LEVEL REQUIREMENTS.....</u>	<u>4</u>
2.1.	PPUK BUSINESS FUNCTIONAL REQUIREMENTS	4
2.1.1.	THE LEGACY PLATFORM AND ITS CHALLENGES	4
2.1.2.	FUNCTIONAL BUSINESS REQUIREMENTS.....	5
2.2.	NON-FUNCTIONAL REQUIREMENTS	7
<u>3.</u>	<u>PPUK AS-IS DATA PLATFORM CAPABILITIES</u>	<u>13</u>
<u>4.</u>	<u>DATA TOOLKIT SERVICES REQUIRED FOR PHASE 1,2 & 3 DELIVERY</u>	<u>17</u>
<u>5.</u>	<u>PHASE 1 – PPUK DATA PLATFORM CAPABILITIES & DATA TOOLKIT MAPPING</u>	<u>20</u>
5.1.	AZURE SQL MANAGED INSTANCE	20
5.2.	AZURE DATA FACTORY SSIS INTEGRATION RUN-TIME.....	21
5.3.	AZURE DEVOPS PIPELINES	21
<u>6.</u>	<u>PHASE 2- PPUK DATA PLATFORM CAPABILITIES & DATA TOOLKIT MAPPING ERROR!</u>	
	<u>BOOKMARK NOT DEFINED.</u>	
6.1.	AZURE DATA FACTORY	22
6.2.	AZURE MACHINE LEARNING	23
6.3.	AZURE DATABRICKS (OPTIONAL)	25
<u>7.</u>	<u>DATA TOOLKIT- DEPLOYMENT BEST PRACTICES</u>	<u>28</u>
7.1.	SERVICE NAMING CONVENTION	28
7.2.	DATA TOOLKIT DEPLOYMENT BEST PRACTICES.....	29

1. Overview

The PPG Data Strategy envisions transforming PPG into a data-driven powerhouse, achieved through strategic data platforms that leverage data as a strategic asset to innovate, optimize, and outperform the market, across the group. Its objectives include establishing and nurturing a data-driven culture, generating actionable insights, driving the adoption of advanced analytics, fostering innovation through data, and ensuring ethical and compliant data practices.

The capabilities of the *Calandra Asset Data Platform* are built using the *Calandra Data Toolkit*, which is deployed in the *Tier 1 Application Landing Zone*. The individual asset capabilities are expected to be delivered in multiple phases.

To comply with local regulatory policies and data security requirements each asset will have its own data platform, and it is intended that data for each asset is combined in the *Transversal Data Platform (Group Asset)* to provide group-wide data and analytics capabilities, and single-source of truth capabilities such as reference data management. Whilst this capability is supported by the platform, by default it is out of scope for initial deployment unless specifically required by the asset.

2. High Level Requirements

2.1. PPUK Business Functional Requirements

2.1.1. The Legacy Platform and its Challenges

Pinnacle Pet Group UK (PPUK) currently operates its data platform using an infrastructure as a service (IaaS) offering on Microsoft Azure, using hosted via virtual machines running SQL Server technology. Whilst technically cloud-based, this configuration is rooted in a legacy, model that no longer meets the demands of a modern, data-driven organisation. It has become increasingly difficult to attract new data professionals, while existing team members express growing dissatisfaction with outdated tooling, creating both recruitment challenges and potential retention risks.

Operationally, the platform is becoming harder to maintain, with mounting complexity and administrative overheads. As data volumes grow, fuelled by continued business success and a projected policy count exceeding two million by the end of 2025, platform performance is deteriorating, particularly in relation to data analysis and query responsiveness. What is more the data platform was based on an existing design, architected with banking systems in mind. This has led to a high-cost, over-engineered solution requiring duplicate server infrastructure across multiple

environments. PPUK now believes these configurations could be safely consolidated within a modern platform-as-a-service (PaaS) model

2.1.2. Functional Business Requirements

The proposed data platform modernisation must meet the evolving needs of a range of senior stakeholders, each of whom has specific functional requirements that cannot be met effectively within the current architecture;

- The **Chief Pricing and Underwriting Officer** requires fast, accurate and auditable data flows to support pricing and risk assessment. This includes the ability to develop and deploy models more quickly, maintain compliance with FCA and PRA regulations, and ensure seamless integration with external systems such as Radar, Polaris and PPUK's Advanced Analytics Environment (AAE). The provision of a high-volume, fully integrated sandbox environment is also essential to enable ongoing research and development.
- The **Head of Data Science** is seeking a data platform capable of ingesting and storing data in multiple formats, including structured data such as tables, flat-files and spreadsheets and unstructured data including PDF documents, satellite imagery and audio files. It is crucial that this is done in a secure and compliant manner. Equally important is the ability to access this data from across the analytics ecosystem and ensure that models developed in Radar and by the in-house AAE team have direct, governed access to the organisation's full data estate. This will enable future use cases to be developed around advanced pricing, behavioural forecasting, and fraud detection.
- The **Head of Analytics** requires an auto-scaling platform that removes the need to manage infrastructure and allows rapid responses to repeatable and ad-hoc queries. A modern 'data lakehouse' architecture is required to unify structured and unstructured data assets, eliminate access bottlenecks, and support real-time streaming as a future capability. Enabling flexible, anonymised reporting views is a key priority, allowing business teams to derive insights more quickly and with greater confidence.
- The **Head of Data** also emphasises the need for a scalable, secure, and accessible platform, with integrated data governance, lineage tracking, and KPI ownership frameworks. Compatibility with legacy source systems, such as the policy management platform Pippa (OpenEdge), must be retained to ensure continuity during the transition. There is also a desire to consolidate infrastructure and simplify the current 'over-engineered' technology stack. Retaining the organisation's existing SQL Server skillsets is a top priority, whilst looking to embrace modern cloud technologies.

- The PPG **Group Chief Data Officer (CDO)** expects PPUK's data platform to be brought in line with modern architectural standards, in support of strategic goals around growth, efficiency, and compliance set at group level. This includes enabling real-time insights, facilitating advanced analytics across all business functions, and supporting referential data management shared across the group to improve data alignment across the UK and EU businesses. Regulatory posture will be strengthened through improved data governance, enhanced reporting accuracy, and a reduced risk of compliance breaches.

Summarising these, we have the following functional business requirements for the modernised PPUK Data Platform;

- **Scalable, cloud-native architecture:** A PaaS platform capable of auto-scaling that supports growth, improves performance, and reduces operational overhead by 15-20%.
- **Unified access to all data:** A central Lakehouse architecture enabling governed access to structured and unstructured data across the organisation.
- **Integration with key systems:** Seamless integration, supporting both real-time and batch ingestion patterns, with platforms such as Radar, Polaris, AAE, and legacy systems (e.g. Pippa).
- **Trusted, high-quality data flows:** Provision of fast, accurate, and auditable data delivery to support pricing, risk, and regulatory compliance (FCA, PRA).
- **Advanced analytics enablement:** The platform should enable support for diverse data types, scalable R&D environments, and direct access for data science and AI workloads developed by the AAE function.
- **Robust governance and control:** Embedded data governance, lineage tracking, and KPI ownership to ensure consistency and compliance.

2.2. Non-Functional Requirements

Ref	Focus	Response
NFR 1	Service Description	
NFR 1.1	Definition	1.1.2 The PPUK Tier 1 Data Platform application is designed to support data outcomes for analytics, reporting, innovation, and data science. 1.1.3 For the purposes of the NFRs outlined here, the following components are in scope: - Data Ingestion (Batch, Intra-day and Event Based) - Data Transformations - Data Storage - DevOps/DataOps tools - Azure Cloud platform - Data governance - Query & visualisation - Machine Learning 1.1.3 Some of the above components are new, whilst others are building on existing capability.
NFR 1.2	Criticality	1.2.1 The service has a medium criticality for RTO (Recovery Time Objective) 1.2.2 If service is not available, it will not impact PPG's ability to trade. 1.2.3 PPUK's current service level agreement (SLA) is 24 hours for core services and 7 days for non-critical services. 1.2.4 Service failure Impact: - Platform data may not be up-to-date for regular reporting - Business Intelligence and Reporting will not be accessible - Analytics Sandbox (Lab) will not be accessible
NFR 2	Usability	
NFR 2.1	Accessibility	2.1.1 System shall support self-service for wide range of the business users. 2.1.2 System shall be reachable from both PPG on-premises and cloud locations. 2.1.3 System shall support PPG IGA solution (Microsoft Azure Entra) RBAC access to include group-wide security policies for System administrators, developers, Analysts with Lab access and read only consumers. 2.1.4 Access to data shall be restricted through the access/permission control in the system's Admin Centre. Each user controlled by a user role matrix that clearly shows the relationships between roles and business functions

		available. The matrix is only configurable to authorised administrators/super users and this allows them to assign security levels, add and delete users, and amend hierarchies and user access rights. An RBAC model ensures that users only have access to information, functions and facilities that are appropriate to their needs and permissions. 2.1.5 The Tier 1 Data Platform application shall support single sign on using Azure Entra ID. SDP shall operate in mixed mode authentication allowing users to sign in via SSO if they have the appropriate privileges, but also allowing suppliers to authenticate via a standard username/password where appropriate. 2.1.6 The Tier 1 Data Platform application shall ensure data encryption at rest (AES-256) and in transit (TLS 1.2+) by default.
NFR 2.2	Volumes - users / data / transactions	2.2.1. An instance of the Tier 1 Data Platform application shall be able to handle up to 1000 concurrent users accessing reporting via PowerBI, Microsoft Excel (via Power Query) and SQL Server Management Services (SSMS) specifically. 2.2.2 The platform shall be able to handle 100 concurrent users for interactive data analytics using Databricks compute. 2.2.3 The platform shall be able to handle 100 concurrent users for interactive data science and analytics using Azure ML compute. 2.2.4. The platform shall be able to accommodate up to 100TB of curated data in lakehouse. 2.2.5. The data platform's Data Lake shall be able to accommodate source raw data in line with PPG group data retention policies, with the potential to store Petabytes of data. 2.2.6 Platform shall be able to scale without requiring physical hardware upgrade. 2.2.7 The data platform shall be able to load transactional data at varying frequency, including batch, intra-day, and event driven (pull).
NFR 2.7	Scalability	2.7.1 The system shall be capable of auto scaling based on the demand placed on the services. 2.7.2 The system shall allow for the configuration of rules that determine the amount of scalability in order to manage cost. 2.7.3 Storage shall be able to scale separately from compute for Data Lakehouse components (Azure Databricks).

		2.7.4 Data ingestion and data transformations in the platform shall be able to scale to different volumes/inbound records.
NFR 2.8	Licensing	2.8.1 The system shall provide interfaces to understand platform spend at the lowest granular level. 2.8.2 Operations teams shall be able to see alerts on resource consumption and be notified when certain limits are reached. 2.8.3 The system shall offer the ability to optimise the cost of different services.
NFR 2.9	Environments	2.9.1 The system shall have segregation of development/test/innovation and any production environments available.
NFR 2.10	Data Retention & Archival	2.10.1 Data retention shall be applied to the system in line with the Group Data Retention policy, as far as is possible.
NFR 3	Reliability	
NFR 3.1	Availability	3.1.1 The system shall target 99.9% (8h41m Yearly downtime) availability/uptime. 3.1.2 Software deployments/upgrades shall happen transparently in the background, users experience no downtime or disruption of service, and are always assured of running on the most-recent release with access to the latest features.
NFR 3.2	Recoverability	3.2.1 The system shall have built-in fault tolerance capabilities. 3.2.2 The system shall make automated backups of databases. 3.2.3 Backups shall be available for the last 7 days 3.2.4 System backups shall be taken once a day 3.2.5 System backups shall be able to be stored in a storage account in a different region from the primary hosting region 3.2.6 There is no requirement for an active disaster recovery environment 3.2.7 In the event of a DR scenario, the data platform shall be able to be returned to functionality with the instance's ITDR and BCP RTO and RTP definitions. 3.2.8 All system code shall be checked into a code repository and version controlled
NFR 4	Performance	

NFR 4.1	Response Time	4.1.1 Queries involving table joins between 2-3 tables to return latest data for current business date within 2 seconds. 4.1.2 The system shall support long-running queries gracefully, and without timing out 4.1.3 System shall be able to respond to operational reports over Bronze Layer within 3 seconds when need arises. 4.1.4 Aforementioned statistics are in relation to the Phase 3 target Data Lakehouse architecture. During the migration of phases 1 and 2, query performance time should be similar to the current 'As-Is' platform. Refactoring of code as part of Phase 1 and 2 migration will be at the discretion of PPUK.
NFR 4.2	Shutdown Time	4.2.1 The system shall support automated and scheduled shutdowns when it is not in use, specifically for services that are pay-as-you go and charge for uptime.
NFR 4.3	Start-up Time	4.3.1 The system shall support automated, event driven and scheduled starts ups, specifically for services that are pay-as-you go and charge for uptime.
NFR 4.4	Throughput - speed of handling above volumes	4.4.1 Ingestion routines must be able to scale to ingest 15 million records per batch. 4.4.2 The system shall support full refreshes and Day 0 loads which process all historic data 4.4.3 The system shall be able to process and transform an incremental load of up to 15 million records per batch within agreed SLAs
NFR 5	Supportability	
NFR 5.1	System Management	5.1.1 All components shall be able to produce and store detailed logs of covering information, warning and error related data 5.1.2 System logs shall support a retention policy to remove or archive them after a set period of time 5.1.3 The system shall provide a reporting interface to assess system health and performance 5.1.4 The system shall be able to ship all, or relevant logs, to Strategic PPG Calandra Sentinel SIEM, managed by a SOC
NFR 5.2	System maintenance / maintainability / upgradability	5.2.1 System upgrades shall be managed by the software vendors with automated patching in place 5.2.2 Systems shall not fall more than one month behind the current release version or software patches for a major upgrade, for minor upgrades system shall not fall more than n-1 version.
NFR 5.3	Antivirus and patching	5.3.1 Antivirus capabilities shall be applied in line with standard PPG policies

NFR 5.4	Releases	5.4.1 All releases shall be able to be rolled back to the previous version
NFR 6	Implementation	
NFR 6.1	Third-party / external components	6.1.1 The system shall provide the ability to manage the spend associated with specific services, such as setting limits to scalability to minimise unexpected spend
NFR 6.2	Ease / approach	6.2.1 The system shall be provided as a service where possible to minimise deployment and installation steps 6.2.2 Where possible, the system shall allow the definition of infrastructure as code so services can be re-deployed via a script 6.2.3 All IaC created shall be genericised, where possible, so the system, or parts of the system, can be easily implemented for other purposes
NFR 6.3	Standards compliance	6.3.1 Development on the platform shall be delivered in line with Calandra development practices.
NFR 6.4	Data toolkit Implementation languages	6.4.1 The system shall support R, Python, Powershell and SQL coding languages for data engineering, data analysis and data science workloads.
NFR 7	Interface	
NFR 7.1	External Systems	7.1.1 The system must be application agnostic and interface or integrate with a range of tools including ETL/ELT, reporting tools, data cataloguing and data science tools. The system must support a wide range of industry standard tools and allow the scope for the addition of future integrations. 7.1.2. System shall support connectivity with other applications via JDBC/ODBC connector and third-party connectors.
NFR 8	Physical	
NFR 8.1	Location	8.1.2 The data platform shall be deployed in the Tier 1 Calandra landing zone, relevant to the business or legal entity consuming the capabilities
NFR 9	ETL/ELT/Streaming	
NFR 9.1	Data Ingestion	9.1.1 System shall be able to ingest data from any source systems via scheduled batch (including intra-day) and/or event-driven (pull). 9.1.2 The system shall be able to support the following types of ingestion along the target cadences as follows; - o Batch – Once per 24 hour window - o Intra-day – At least every 6 hours, at best every 15 minutes - o Event Driven (Pull) – 0.5 to 5 seconds frequency

		9.1.3 System shall be able to transform the data as per the agreed Tier 1 data platform application data model in both Batch and intraday batches.
NFR 10	Data Governance	
NFR 10.1	Data Dictionary & Data Lineage	10.1.1 System shall be able to deliver data architecture documentation and a data dictionary (using the toolkit) to ensure the data model is understood across all assets and easily shared and used for collaboration. 10.1.2 The data platform shall provide complete data lineage to comply with PPG group data auditing policies. 10.1.3 The data platform shall enable the enforcement of data ownership within each Tier 1 asset, including the ownership of consistent key performance indicator (KPI) definitions.
NFR 10.2	Compliance	10.2.1 The data platform must comply completely with prudential regulation authority (PRA), financial conduct authority (FCA) and the general data protection regulations (GDPR).
NFR 10.3	Data Management	10.3.1 The data platform shall enable sharing of referential data management (RDM) across Tier 1 assets in the PPG Group.
NFR 11	laC	
NFR 11.1	Infrastructure as Code(laC)	11.1.1 Data toolkit shall be deployed as per the Calandra's Infrastructure as Code (laC) standards.

3. PPUK As-Is Data Platform Capabilities

This section provides a review of the PPUK data platform's 'As-Is' capabilities. The system context diagram, shown in Figure 1, outlines the core components, data flows, and integration touchpoints of Pinnacle Pet Group UK's (PPUK) existing data platform. It depicts how internal and external systems contribute to, and consume from, the central SQL Server-based data platform, including analytics tooling and stakeholder access mechanisms.

The **PPUK Data Platform** is built upon a traditional *Microsoft SQL Server Data Warehouse* architecture. SQL Server is installed on an Azure hosted VM, managed by the PPUK team as IaaS. The architecture includes two VM's hosting SQL server, one handling the back-end data processes and the other the front end (analytic and presentation layer). This is now being consolidated and the current as-Is design reflects the to-be single server architecture.

The vast majority of data sources operate on a *batch-driven* ingestion model, with approximately 90% of data arriving via *daily loads*. There are notable exceptions, such as the source system providing quote data, which delivers *data hourly* via an extracted XML and is monitored by a *15-minute intraday service* to ensure quote integrity. The *policy management system (Pippa)* uses *Change Data Capture* (CDC) and is *event-driven*, making it a critical path in the ingestion process. The *on-premises* source systems connect *via SQL Server linked services* and are orchestrated through *SQL Agent Jobs*. These jobs are typically serialised rather than parallelised, and ingestion pipelines load into individual *Extract databases* for each source, *retaining seven days of data*.

From a storage perspective, typical daily volumes are up to 15 million rows of raw data ingested into the platform. Of these 15 million, 5 million approximately end up as changes made in the *Integrated Data Warehouse (IDW)* after data cleansing, transformation and conforming to the data model. The IDW maintains active tables for the *current month-to-date* and *three full months* of data. RAW and staging tables are truncated at the start of each ETL cycle, and backups are taken daily at the end of ETL processing. These include the *IDW* and *Datamart Aggregates* databases. Incremental backups are taken during the week, with full backups at the weekend and post-release. The *recovery SLA* is *24 hours* for core systems and up to *seven days* for *non-critical reporting*.

The orchestration and transformation layer is entirely managed via *SSIS packages*, with execution controlled through *SQL Agent Jobs* and, where needed, embedded SQL or PowerShell scripts. See the expanded *Data Platform Server* view in Figure 2. A metadata-driven approach underpins the orchestration logic, managed through a database called *Information Management*. Although the platform will shortly reside on a single VM there is still separation between the, so-called, 'back-end' processes and the 'front-end' presentation layer. The *IDW* is replicated using a *SQL Server Read-Only replication*

mechanism. Approximately 95% of IDW tables are replicated, with some exclusions for high-volume tables. Once the platform has migrated on to a single VM, which is expected to be a prerequisite for the Phase 1 migration, the replication of the IDW will be deprecated and only a single IDW database will remain.

The IDW serves as the foundation for the presentation layer. This also houses a ***Datamart Aggregates database***, which provides the semantic model and reference data for business reporting. SSIS flows, triggered in completion of the 'back-end' processes, trigger the population of these structures, with the *Information Management* instance governing the metadata driven orchestration for the presentation layer. ***Power BI*** is the primary visualisation tool, with connectivity pointing to the Data Marts aggregate database. Currently, the team uses a Microsoft Fabric ***Power BI SaaS instance***, with report refreshes scheduled using Power Automate. ***Internal data access*** is also facilitated via ***Excel Power Query***, ***SAS*** and ***Python*** for workflow automation. There is also a LABS environment (sandbox) for SQL experimentation and the testing of new business aggregate views. Data exports to ***external partners*** are handled via ***flat file generation in SSIS***, written to server directories and then ***transferred via SFTP***.

Code control is robust, with all SSIS and SQL code managed through ***Azure DevOps***. While there is no CI/CD automation in place yet, DevOps is used for source control and release management. Releases are ***manually triggered*** using ***PowerShell scripts***, based on sprint-defined release branches compared to the main branch. These scripts are deployed manually to the server, and then the team remotes into the environment to execute deployments.

Data governance is partly implemented. Authentication is managed via ***Microsoft Entra ID*** (formerly Azure AD), with ***role-based access control (RBAC)*** used to govern permissions. While access to RAW data is typically read-only, some analysts do have direct access for specific use-cases. There is no formal data catalogue, though ad-hoc Excel sheets are used to track schema and lineage.

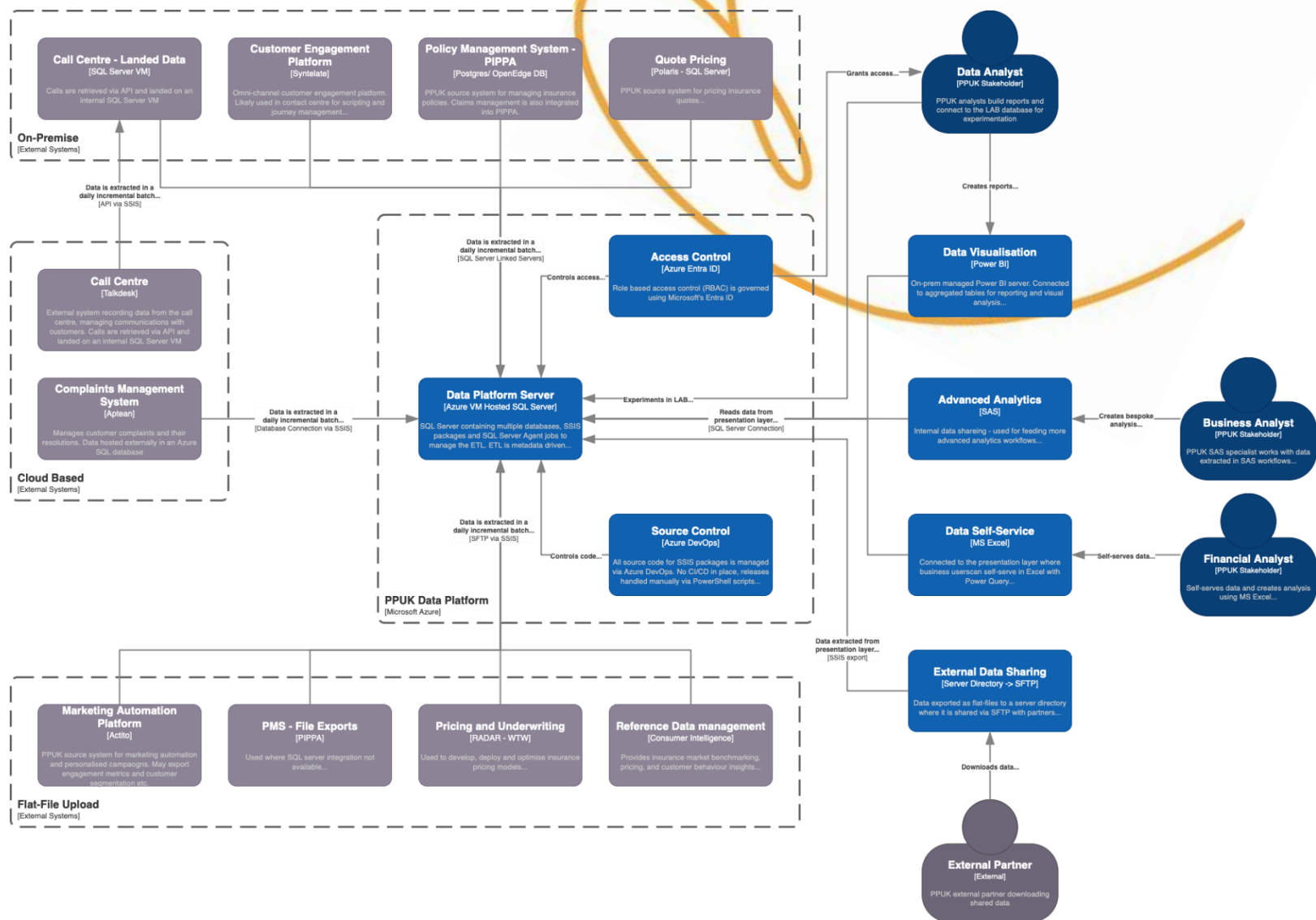


Figure 1: System Context view of current PPUK 'As-is' data platform landscape

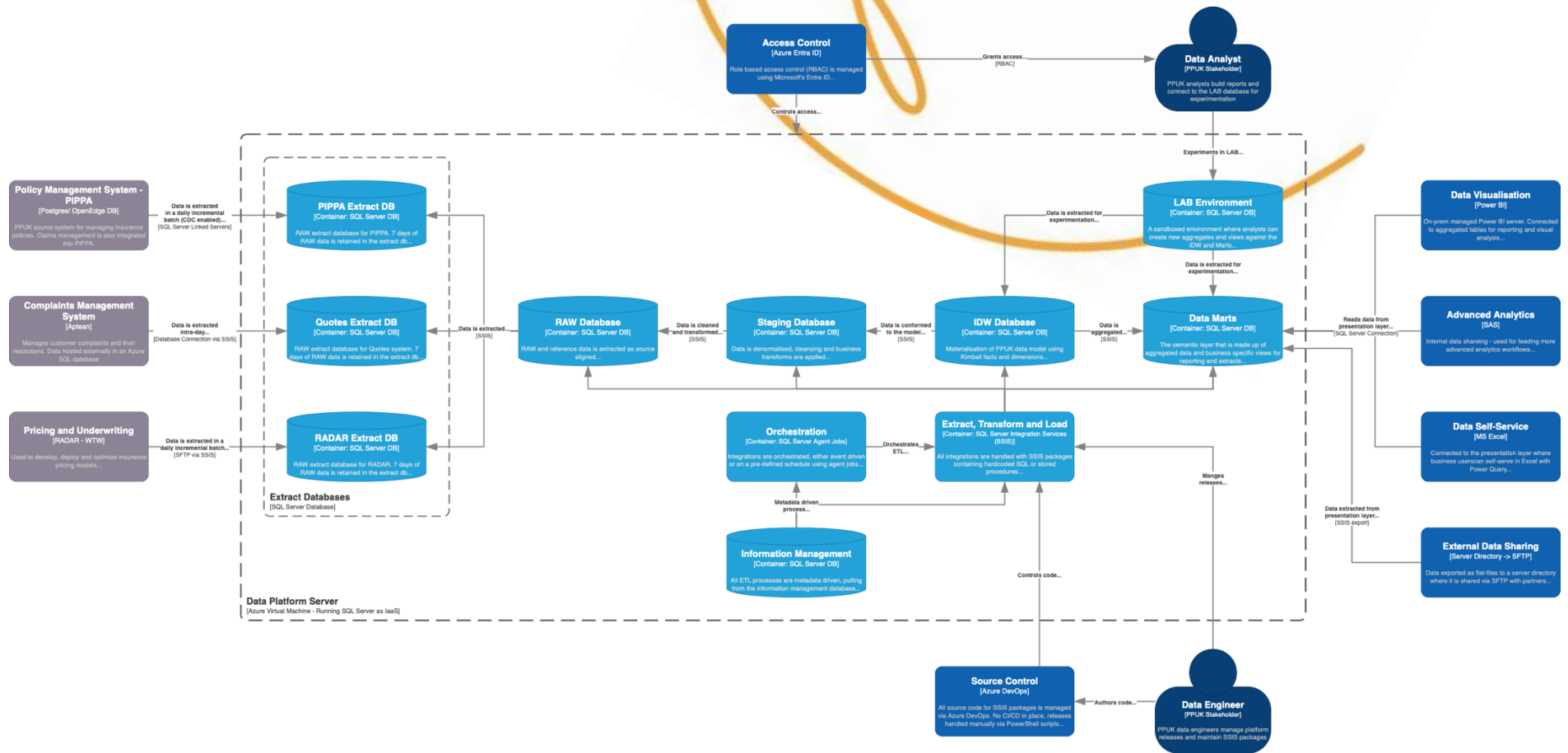


Figure 2: A System Container (C2) level diagram of the Data Platform Server

4. Data Toolkit Services Required for Phases 1 and 2 Delivery

The plan is to carry out the migration of the current PPUK 'As-Is' data platform architecture across three phases. This approach is designed to minimise disruption whilst ensuring a continuous modernisation of the data platform, moving towards the PPG group target data platform architecture:

Phase	Description	Objectives
Phase 1	The first phase of PPUK's data platform re-design is split into two parts; Part A.) Migrate the existing 'As-Is' SQL Server VM based architecture to Azure SQL Server Managed Instances. Migrate existing SSIS packages to ADF using the SSIS integration run-time. Separate ETL processing for 'back-end' and 'front-end' processes. Part B.) Begin process of migrating away from legacy SSIS workloads, moving packages into native ADF pipelines. Build support for AAE and RADAR integration with the Data Platform	The objectives are aligned as follows; Part A.) The primary objective of Phase 1 is to migrate to the use of PaaS instances, which will lead to reduce operating costs and improved redundancy. Offer an improved end-user experience (access to data, query performance and report performance). Part B.) The goal of Phase 2 is to completely migrate away from using legacy SSIS packages in the platform. It will also provide data connectivity for the AAE team and deliver RADAR integration with the platform.
Phase 2	Review data platform architecture in relation to the wider PPG group architecture principles, aligned closely to the Data Toolkit. This will include embracing the wider use of Azure Databricks for supporting real-time data workflows, and Azure Data Lake Gen 2 for flexible, high volume data storage	The goal of Phase 3 is to bring the PPUK data platform in line with the approved PPG group data platform architecture, utilising services specified in the Data Toolkit.

The elements of the Data Toolkit required for each phase are outlined below. Phase 1 will seek to deploy and stand-up all core services required to achieve the objectives, with Phase 2 services reserved for fulfilling future use cases:

Data Toolkit	Purpose	Phase 1	Phase 2
Azure Data Factory (ADF)	Data integration service for creating data-driven workflows to move and transform data. Purpose includes data movement, ETL, and orchestration.	X	
Azure Data Factory (ADF) – SSIS IR	The SSIS integration run-time implementation of ADF. Supports legacy SSIS packages, executing from an SSISDB hosted on a SQL Server Managed Instance	X	
Azure Function App	Serverless compute service for running event-driven functions. It allows scalable execution of code with minimal infrastructure management.	X	
Azure Key Vault	Securely manages secrets, keys, and certificates. It centralizes access to sensitive information for applications and services.	X	
Azure Databricks	Unified analytics platform for big data and AI workloads. It provides Apache Spark-based environments for data engineering and data science tasks.	X	
Azure Storage Accounts	Cloud storage solution for storing data, with different types of storage options (Blob, File, ADLS Gen2) for data management and access.	X	
Azure SQL Managed Instance (MI)	Managed SQL Server instance that replicates much of the capability of a traditional SQL Server IaaS deployment. Includes support for SQL Agent Jobs, SQL Linked Servers and cross-database querying.	X	
Azure Event Hub	Real-time data streaming platform for collecting, processing, and storing event-driven data streams at massive scale.		X
Azure Event Grid	Event routing service that enables serverless event-driven architectures by delivering events from various sources to Azure services.		X
Azure Purview	Unified data governance service to manage and protect data assets across your entire organization. It provides data cataloging and lineage capabilities.	X	
Azure DevOps	Development and collaboration platform for DevOps practices, including CI/CD pipelines, version control, and project management.	X	
Power BI	Business analytics service to visualize and share insights from data through interactive reports and dashboards. Allows creation of interactive reports and dashboards,	X	

Data Toolkit	Purpose	Phase 1	Phase 2
	connects to a wide range of data sources, performs data transformation and analysis, and shares insights within organizations.		
Azure Machine Learning (Azure ML)	A cloud-based machine learning platform to build, train, and deploy machine learning models. Provides tools for building, training, and deploying machine learning models. Supports automated ML, model versioning, experiment tracking, and integration with frameworks like TensorFlow, PyTorch, and Scikit-learn. Also offers model deployment and monitoring in production environments.	X	

5. Phase 1 – PPUK Data Platform Individual Capabilities & Data toolkit Mapping

The Tier 1 Data Platform application's capabilities are built on the Microsoft Azure Cloud stack. These capabilities are expected to be implemented incrementally according to the prioritised requirements.

The phase 1 design detailed here is based on the objectives set out under section 4;

1. To modernise the current PPUK As-Is data platform architecture, moving from deployed infrastructure as a service (IaaS) to platform as a service (PaaS) or software as a service (SaaS) offerings.
2. To reduce operational overheads.
3. To improve end-user experience through improved query and report performance.
4. Enable secure connectivity into the platform for the Advanced Analytics Environment (AAE).
5. Stand-up and enable a Databricks LAB environment for PPUK team internal experimentation and development

The design remains largely in-line with the As-Is architecture shown in Figure 2, particularly in terms of database structure, however there are a number of key elements that have been included to support the objectives for phase 1.

5.1. Azure SQL Managed Instance

The design centres around moving away from the current IaaS architecture to PaaS, using an Azure SQL Server Managed Instance (MI) in place of SQL Server running from an Azure hosted virtual machine. The managed instance offers a number of key benefits over the IaaS approach;

- The fully managed platform requires ***no OS level patching, backups or maintenance tasks*** – this is all handled by Azure.
- As a PaaS service the MI offers ***high availability, automatic backups*** and ***point in time restore*** up to 35 days.
- SQL Server MI offers ***near 100% compatibility*** with traditional SQL Server, including the use of ***SQL Agent, Linked Servers*** and ***cross-database*** querying
- Offers an efficient route ***to performance scaling***, with ***compute*** and ***storage*** able to be ***scaled independently*** with minimal downtime.

- The SQL Server MI benefits from the **Azure Hybrid Benefit** for SQL Server licensing, potentially lowering costs.

A core additional database that will need to be deployed on the Azure SQL MI is the package store and logging database for the Azure Data Factory SSIS integration run-time, SSISDB. This is a core component, and can be shared between multiple integration runtimes.

5.2. Azure Data Factory SSIS Integration Run-time

To migrate existing SSIS functionality we use Azure Data Factory (ADF) instances with the SSIS integration run-time (SSIS IR). Whilst SSIS is now considered a legacy service, with Microsoft discontinuing future development and focusing on solely on security, it is widely used and much of the functionality can be achieved with the ADF SSIS IR.

ADF SSIS IR execute packages that are deployed to the SSISDB, hosted on the SQL Server MI. Best practice here would be to wrap SSIS packages handling specific workflows into a single master package that is then executed by the ADF SSIS IR. Orchestration can either be managed by an ADF Pipeline by executing the *'Execute SSIS packages'* activity OR via the use of SQL Agent Jobs on the MI.

In the phase 1 design it is intended to orchestrate the execution of SSIS packages using SQL Agent Jobs, as is the current approach. This will minimise disruption and ensure a smooth migration.

The phase 1 design calls for the deployment of two ADF SSIS IR instances. One will manage the 'back-end' processes and the other the 'front-end'. This will enable us to achieve one of our core objectives of improving end-user experience by separating the workloads. Both these instance will execute packages and log executions from and into the same common SSISDB hosted on the SQL Server MI.

5.3. Azure DevOps Pipelines

One of the core pieces of functionality not supported by the SQL Server MI is the ability to execute PowerShell scripts and deploy packages in managed releases, as is possible with the current IaaS approach. There are two approaches that we might consider to replicate this functionality with the SQL Server MI;

1. **Azure DevOps Pipelines Approach** – This would be the recommended approach, relying only on Microsoft-hosted agents, however represents a move away from the current supported approach. It involves;
 - a. Building the *'ispac'* file from the SSIS *'.sln'* file using the supported **MSBuild** command.

- b. Once built, use T-SQL and SSISDB stored procedures to deploy the package to SSISDB. These can be wrapped in a PowerShell script and is invoked in the Azure DevOps Pipeline with the 'Invoke-SqlCmd' step.
2. ***Replicate the current process by provisioning a dedicated 'Release Management' VM.*** – This approach is designed to replicate closely what is done today, which PPUK engineers connecting to a VM to release packages. As this environment is considered 'self-hosted' there is more flexibility in how packages can be deployed;
 - a. Using the 'ISDeploymentWizard' executable
 - b. Using the 'Deploy-ISProject' PowerShell Cmdlet

In both approaches packages are deployed to separate folders in the SSISDB on the SQL Server MI and these are executed by the ADF SSIS IR.

Although it will require a move away from the current as-is way of working, it is recommended that approach 1 is adopted by PPUK. This approach aligns with the phase 1 objectives to move all services away from IaaS. It also takes a step further in modernising the PPUK platform, with implementing continuous integration and deployment (CI/CD) possible in this approach

The aforementioned Azure services are required to enable the delivery of the migration objectives under Part A. The design of the Part A architecture is shown in Figure 3. The services described in the following sub-sections are required to enable the delivery of the objectives under Part B.

5.4. Azure Data Factory

In Phase 2 the data platform will seek to retire legacy SSIS packages and transition all data ingestion and transformation activity into **Azure Data Factory (ADF)**. ADF provides a modern, cloud-native orchestration layer that is fully managed, scalable, and integrated with the Azure security and monitoring stack. This ensures a sustainable long-term approach, aligned with Microsoft's direction of travel for data integration.

As per the phase 1 deployment, the phase 2 design will support the following ingestion mechanisms

5.4.1. Database Sources (Linked Server replacement)

Instead of the Azure SQL legacy *Linked Server* connections, each source database will be defined as a Linked Service in ADF. Where sources are hosted on-premises, the self-hosted integration run-time (SHIR) is installed on the source server and will broker the connection securely. Copy Activities within ADF pipelines will then move

the data directly into the designated SQL MI Extract DB depending on the source system.

5.4.2. API Sources

ADF supports a native HTTP connector, which will be used to extract data from simple RESTful APIs. Pipelines can define authentication headers, query parameters, and pagination rules. In the event that complex patterns are required (e.g. token refresh or pre-processing), lightweight Azure Functions can be invoked within the pipeline. Data is then written into the designated SQL MI Extract DB depending on the source system.

5.4.3. Flat Files via SFTP

ADF's SFTP connector will retrieve files deposited on external servers using key-based authentication. A straight forward Copy Activity is then used to land the data into the designated SQL MI Extract DB depending on the source system. Event triggers or polling can be configured depending on whether ingestion should be scheduled or near real-time.

Once the data is extracted into the Extract DB landing zone, ADF pipelines will then perform the required cleansing and transformation steps by calling stored procedures in each of the target databases.

This configuration ensures that ADF acts as the central, governed ingestion hub, supporting all major integration patterns required by the platform: databases, APIs, and file-based exchange. Azure Key Vault should be used for storing secrets and Azure DevOps should be used to source control the ADF pipelines as code.

5.5. Azure Machine Learning

The Advanced Analytics Environment (AAE) at PPUK is already experienced in the use of the Azure Machine Learning (AML), which provides a secure, scalable, and fully managed environment for data science activity. By using AML workspaces as the front door for advanced analytics, we ensure that experimentation and model development are carried out within a governed Azure service rather than dedicated Azure VM's, unmanaged desktops or local tools. This approach centralises security, provides enterprise-grade monitoring and version control (through tight integration with Azure DevOps), and integrates naturally with the wider Azure ecosystem, including identity, networking, and storage. It also allows Data Scientists to collaborate in a controlled way without compromising the security of the core data platform.

As part of the Phase 2 deployment, the AML workspace will be configured with:

- **Notebook workspaces:** hosted Jupyter notebooks where Data Scientists can write and run Python code.
- **Compute resources:** AML compute instances and clusters will be deployed into a secure VNet-injected subnet, ensuring that all processing occurs within the private Azure network boundary.
- **Managed identity integration:** AML will be configured with system- or user-assigned managed identities so authentication can be performed without exposing credentials and managed via RBAC.

Azure SQL Managed Instance (MI) is deployed without a public endpoint and will be exposed to AML through a **private endpoint** within the virtual network. The AML compute resources will connect over this private channel, ensuring all data access remains within the trusted Azure backbone.

Data Scientists can then query MI *programmatically* within their AML notebooks using Python libraries such as *pyodbc* or *sqlalchemy*. Authentication is handled via the managed identity of the AML compute, which is mapped to an Azure Entra ID login configured in the SQL MI. This approach removes the need for usernames and passwords in code, enforces role-based access control, and provides a fully auditable authentication flow.

5.6. Azure Blob Storage

To support the Advanced Analytics Environment (AAE) and the work of the Data Science team, an **experimental data store** will also be provisioned as part of the Phase 2 design using **Azure Blob Storage**. This will provide a secure, familiar and flexible staging area for data used in model development, experimentation, and reproducibility.

The core design principles for this decision are as follows;

- **Provide Read-only access to the core data platform:** Data Scientists will connect to the SQL Managed Instance via Azure ML workspaces using managed identities with *read-only* privileges. Source data required for experimentation will be extracted once and persisted into Blob Storage to avoid repeated heavy queries against the production system. This is in line c=with current ways of working for AAE, who use dedicated Azure Blob containers per project for storing artefacts.
- **Efficient file formats:** High volume extracted datasets should be stored in *Parquet* format to maximise performance, support schema preservation, and enable efficient reuse in analytical workflows. For smaller datasets .csv flat file or .json is also acceptable.

- **Reproducibility and audit:** Training and testing datasets used to build models will be retained in Blob Storage, as per current AAE ways of working, to allow full reproduction of results. This ensures compliance with audit requirements and provides a clear lineage of model inputs.
- **Cost-effective lifecycle management:** Azure Blob Storage *lifecycle policies* should be configured to automatically transition data from the *Hot* tier to lower-cost storage after a defined retention period. A Recommended lifecycle management approach for the initial AAE use cases is as follows;
 - Active projects, and those that have been completed in the prior 6 month period will remain in *Hot* storage, where data can be accessed instantly
 - After the initial 6 month period data will be transitioned to the 'cool' tier, where access is still guaranteed although it may take minutes to retrieve the data.
 - After two years, the designated PPUK retention policy period, data will be moved to Archive storage for long-term compliance retention. Data access is guaranteed, however time to retrieve is on the order of hours.

This approach balances **security**, **efficiency**, and **cost management**, while aligning with the team's existing workflows and familiar tooling within Blob Storage.

5.7. Azure Databricks (Optional)

As PPUK have had limited exposure to Azure Databricks, and considering this is a core element of the target Clandra Tier 1 data platform application architecture, we recommend considering the deployment of an Azure Databricks workspace to act as a LAB environment for experimentation and exploration. This might include;

- Explore connecting to source systems using Databricks Lakeflow, a collection of managed connections for popular services.
- Explore creation of ETL workflows using Databricks Jobs or Delta Live Tables pipelines.
- Explore data storage and curation with Unity Catalogue and Delta Tables.
- Experiment with querying data and conducting analysis using the Databricks Workbook interface.

The use of Azure Databricks would also necessitate the deployment of an Azure Data Lake Storage (ADLS) Gen2 instance to manage storage of data in Databricks as through Unity Catalogue and delta tables.

The design of the Part B architectures is shown in Figure 4.

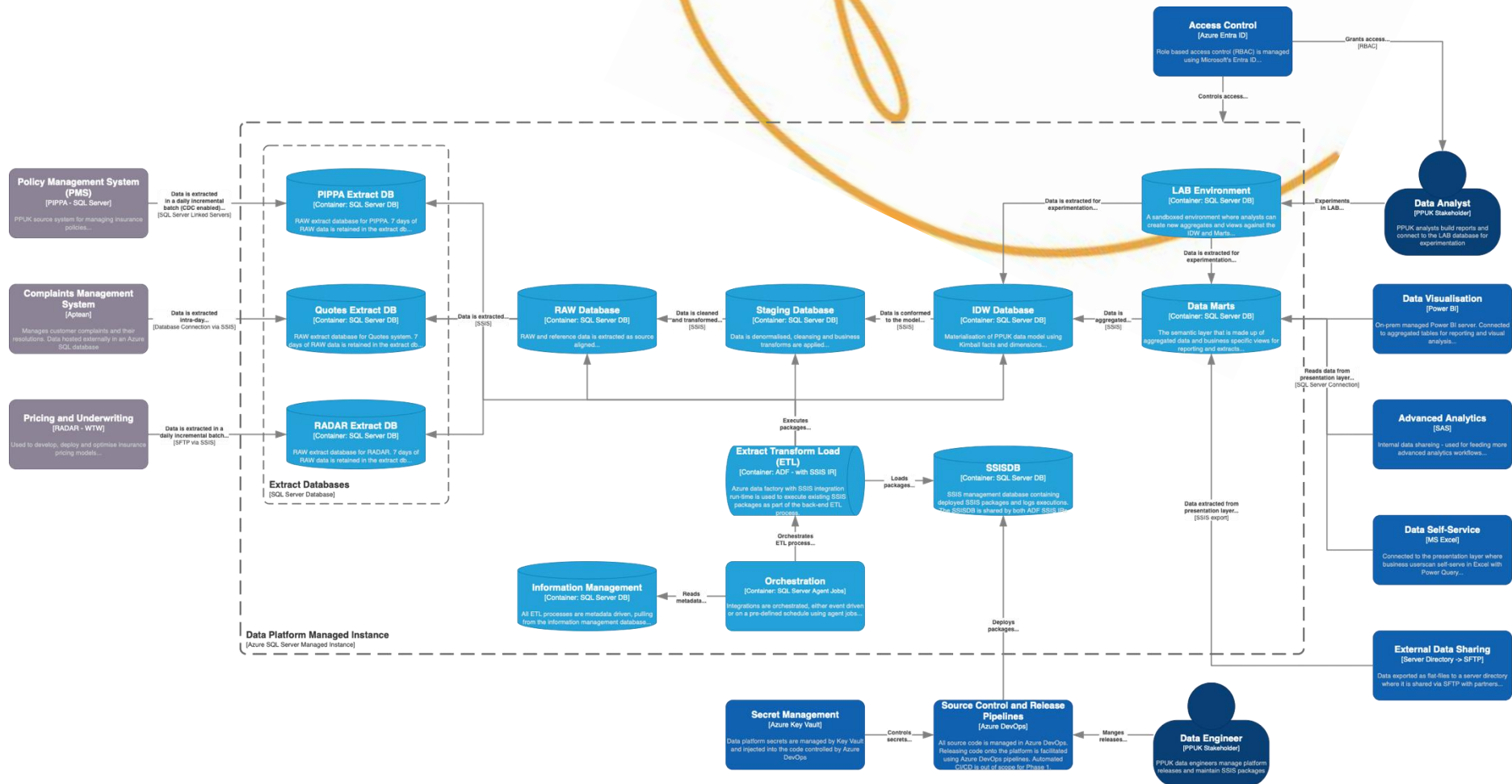


Figure 3: Target architecture for delivery of requirements under Phase 1, Part A

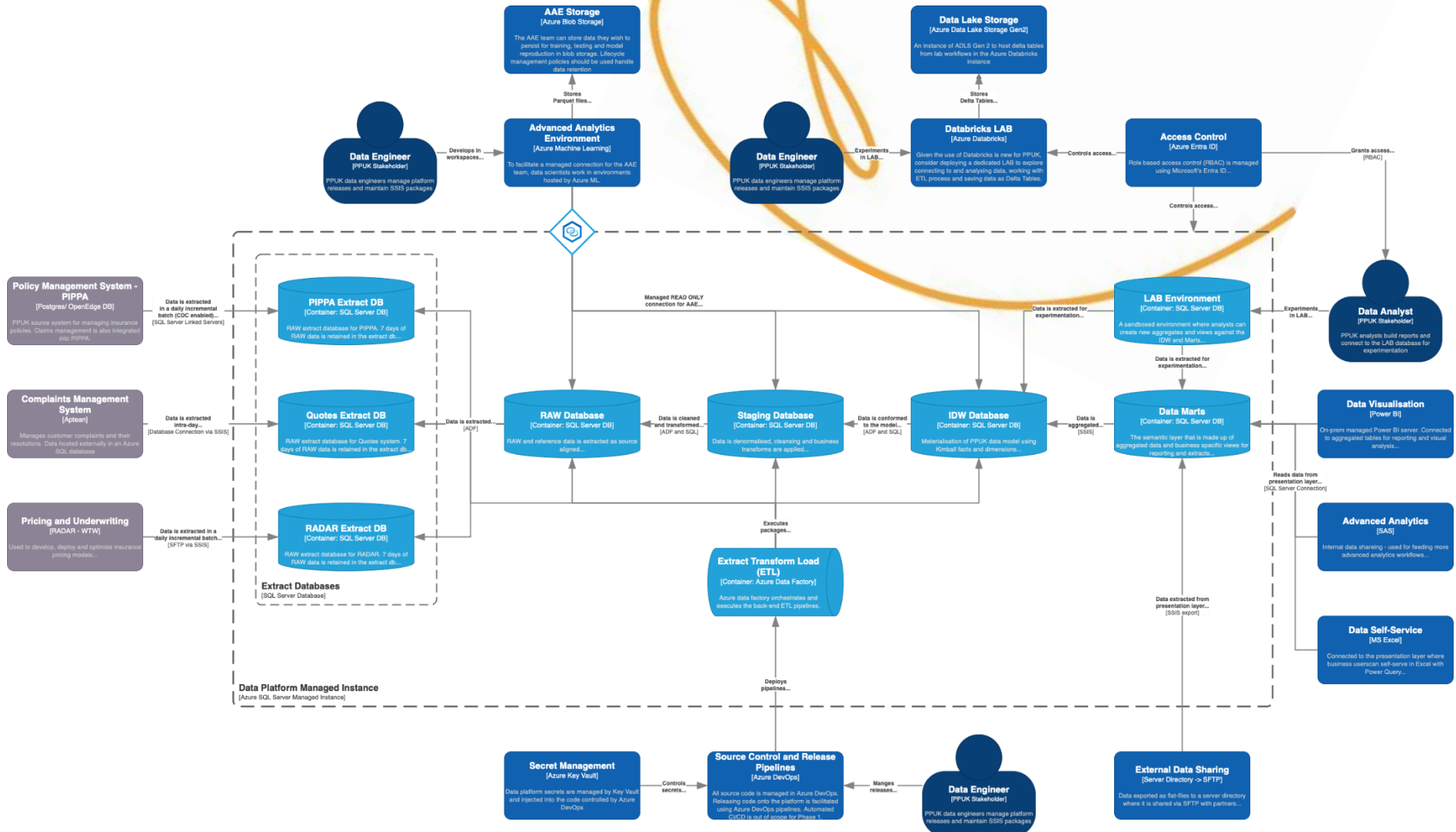


Figure 4: Target architecture for delivery of requirements under Phase 1, Part A

6. Data Toolkit - Deployment Best Practices

This Section outlines data toolkit deployment best practices and high-level configurations of the tools in data toolkit required to deliver Tier 1 Data Platform applications.

6.1. Service Naming Convention

We are employing the service naming convention in-line with that used for the Calandra Azure Landing Zone design. The templated format is as follows;

ppg-<service>-<asset>-<identifier>-<region>-<environment>-<count>

Where;

- <service> is replaced by shorthand notation of the azure service being deployed;
 - e.g. for Azure Data Factory this would be 'adf'
- <asset> is replaced by shorthand notation of the asset where the service is deployed;
 - e.g. for Pinnacle UK this would be 'puk'
- <identifier> is replaced by shorthand notation describing a particular instance of the service;
 - e.g. for example for SSIS integration runtime this might be 'ssis'
- <region> is replaced by shorthand notation for the region where the service is deployed;
 - e.g. for UK South this would be 'uks'
- <environment> is replaced by shorthand notation for the desired environment;
 - e.g. for production this would be 'prod'
- <count> is replaced by the count of identical instances of a service;
 - E.g. If PPUK have two ADF instances with SSIS IR then the count on each will be 001 and 002.

Additional points on the naming convention template;

1. Where possible it is desirable to keep the shorthand notation to three characters for brevity.
2. Where possible use the hyphen '-' character to separate the terms in the name. Where this is not possible (notably in Azure storage accounts) use either the underscore character '_' or have no space between each term.

6.2. Phase 1 - Data Toolkit Deployment Best Practices

Data Toolkit	Deployment Best Practice
Azure Data Factory (ADF) – SSIS Integration Run-Time	• Name: <i>ppg-adf-ppuk-ssis-uks-<environment>-001</i> • Instances: 2 (one for back end, and front-end processes) • Size: e16 V3: 16 Cores, 128 GB RAM and 400GB Disks • Region: UK South • Environments: PPUK requires ADF SSIS IR instances services deployed across the following environments; • Dev • Test • UAT • Pre-Production • Production • Configuration: ○ Integration runtime type: Azure-SSIS IR ○ Node Size and Count: Standard_D2_v3 with single node initially. ○ Enable 'auto-pause' to handle infrequent workloads ○ Edition: STANDARD by default ○ Deploy SSISDB (integration services catalogue) to an Azure SQL Managed Instance. • Networking & Security: ○ Use VNET Integration for secure network access. ○ Enable Managed Identity for Data Factory. • Monitoring: ○ Enable Diagnostic Settings and Application Insights. • Best Practices: ○ Credentials shall be stored in Key Vault ○ A Data Factory Managed Identity must be created for the Data Factory. ○ Managed Virtual Networks and Private links must be used for Azure SSIS Integration Runtime connecting to linked services (on-premises and private endpoints). ○ Separate Data Factory created for each environment. ○ Managed identities for Azure resources shall be used as a preference for authentication to the Key Vault ○ Resource Tagging for costing and billing must be applied according to Calandra Tagging policies. ○ Packages should be deployed into SSISDB and can be configured through the ADF UI or SSMS. ○ SQL Agent Jobs can be configured on a SQL Server managed instance against SSISDB using

Data Toolkit	Deployment Best Practice
	<i>SSISDB.catalog.create_execution</i> and related stored procedures - Set up Azure Monitor alerts on SSIS integration runtime metrics, such as 'IR start/stop' and 'CPU/memory' thresholds, to support proactive monitoring of service performance. - The ADF SSIS integration runtime should be shut down when not in use to reduce costs. This can be done via Azure Automation and/ scheduling in PowerShell.
Azure Key Vault	Name: <i>ppg-akv-ppuk-uks-<environment>-001</i> - Instances: 1 - Region: UK South - Environments: PPUK requires Azure Key Vault services deployed across the following environments; - Dev - Test - UAT - Pre-Production - Production - SKU: Standard (or Premium for HSM-backed keys) - Security & Compliance: - Use RBAC for access management. - Restrict access by enabling VNET integration and firewall rules. - Monitoring: - Enable Diagnostic Settings and Application Insights. - Best Practices: - Maintain separate Key Vault for different environments. - Enable soft delete to allow recovery of deleted vaults and vault objects. - Use service endpoints to connect securely from Vnet to vault - Principles of least privilege and limit who has access to azure resources. - Managed identities for Azure resources shall be used as preference for authentication to Key Vault. - Key Vault keys & secrets shall have an expiration date as per Calandra Landing Zone standards. - Certificates shall have the specified maximum validity period as per Calandra Landing Zone standards.

Data Toolkit	Deployment Best Practice
Azure SQL Managed Instance	• Name: <i>ppg-ass-ppuk-sqlmi-uks-<environment>-001</i> • Instances: 1 • Region: UK South • Environments: PPUK requires SQL Serve Managed Instances deployed across the following environments; • Dev • Test • UAT • Pre-Production • Production • Compute Tier ○ Use General Purpose Tier as standard ○ Select vCore-based purchasing model to align predictable performance with licensing and make use of Azure Hybrid Benefit (if applicable). ○ Starting Configuration: 16 vCore Standard Series (Gen 5). Supporting up to 4TB of storage. ○ Select auto-growth enabled on storage to reduce operational overhead. • Network and Security: ○ Deploy into a dedicated private subnet within a virtual network (VNet). ○ Use private endpoint for secure, private connectivity to Azure services and integration with PPG's broader data platform. ○ Use Azure Defender for SQL to detect threats such as SQL injection and anomalous access patterns. ○ Ensure transparent data encryption (TDE) is enabled by default; use customer-managed keys (CMK) in Azure Key Vault if needed. ○ Enforce TLS 1.2+ for secure transit of data. • Backup and Disaster Recovery: ○ Enable 35 day point in time restore (PITR) ○ Enable long-term retention (LTR) for compliance or audit needs. ○ For production instances enable geo-replication to a paired region (active/passive) ○ For development/ tests instances zonal-replication is sufficient • Monitoring and Alerting: ○ Enable Diagnostic Settings and Application Insights

Data Toolkit	Deployment Best Practice
	○ Use Azure Monitor for SQL Insights to gain deep visibility into performance, availability, and query metrics. ○ Enable SQL Auditing to track database activity and user access. ○ Create alerts for key conditions: ▪ CPU/memory thresholds ▪ Blocking sessions ▪ Long-running queries ▪ Storage nearing limits ● Identity and Access Management: ○ Use Azure Entra ID authentication for centralised user access control and auditing, in line with Calandra group policies. ○ Disable SQL authentication by default unless required for legacy system compatibility. ○ Use Managed Identity for secure service-to-service access (e.g. ADF, Azure Functions connecting to SQL MI). ○ Enable row-level security (RLS) and dynamic data masking where applicable to restrict sensitive data exposure. ● Best Practices; ○ Assess and optimise existing on-prem workloads using DMA (Data Migration Assistant) prior to migration. ○ Use Azure Data Factory or DMS for migrating schema, data, and logins with minimal downtime. ○ Separate environments (Dev/Test/Prod) via different SQL MIs or use multiple databases with agreed naming conventions and RBAC. ○ Implement scheduled index maintenance and statistics updates via SQL Agent jobs or maintenance solutions. ○ Enable Query Store to monitor execution plans and regressions over time. ○ Resource Tagging for costing and billing must be applied in advance as per Calandra Landing zone policies. ○ Run monthly cost/performance reviews using Azure Advisor and Cost Management tools.
Azure Purview	● Name: <i>ppg-apv-ppuk-uks-001</i> ● Instances: 1 ● Region: UK South ● Environments: N/a – 1 purview instance per tenancy allowed

Data Toolkit	Deployment Best Practice
	• Compute SKU: Standard for non-production and Premium SKU for production. • Data Sources: ◦ Register Azure SQL MI for data governance. ◦ Register ADLS Gen2 Account for data governance ◦ Register Azure Blob Storage Account for data governance. • Security & Compliance: ◦ Use Managed Identity for authentication. ◦ Enforce RBAC-based access control. • Monitoring & Logging: ◦ Use built-in monitoring for scanning activities.
Azure DevOps	• Name: <i>ppg-ado-ppuk-uks-001</i> • Instances: 1 • Region: UK South • Environments: N/A • License & Access Control: ◦ Enable the "Basic" License for all users in the ADO project. ◦ Add users to the Project Contributor group. • Repositories & CI/CD Pipelines: ◦ Use Git repositories with branch policies. ◦ Automate deployments using YAML pipelines. • Security & Compliance: ◦ Enable RBAC and Azure DevOps Policies. ◦ Use Service Connections with Managed Identity.

Data Toolkit	Deployment Best Practice
Azure Data Factory (ADF)	• Name: <i>ppg-adf-ppuk-gp-uks-<environment>-001</i> • Instances: 2 (one for back-end and front-end processes) • Region: UK South • Environments: PPUK requires Azure Data Factory instances deployed across the following environments; • Dev • Test • UAT • Pre-Production • Production • Compute Type: General-purpose • Networking & Security: ◦ Use VNET Integration for secure network access. ◦ Enable Managed Identity for Data Factory.

Data Toolkit	Deployment Best Practice
	- Monitoring: - Enable Diagnostic Settings and Application Insights. - Best practices: - Credentials shall be stored in Key Vault - A Data Factory Managed Identity must be created for the Data Factory. - Managed Virtual Networks and Private links must be utilized for Azure Integration Runtime connecting to linked services. - Separate Data Factory created for each environment. - Managed identities for Azure resources shall be used as a preference for authentication to the Key Vault - Resource Tagging for costing and billing must be applied according to Calandra Tagging policies.
Azure Function App	Name: <i>ppg-afa-ppuk-<identifier>-uks-<environment>-001</i> - Instances: 1 – scaffolding deployment in readiness for future use cases - Region: UK South - Environments: PPUK requires Azure Functions deployed across the following environments; - Dev - Test - UAT - Pre-Production - Production - Hosting Plan: - Consumption (standard) Plan for production workloads. - Consumption (standard) Plan for non-production workloads. - NB: Switch to Premium plan for production workloads only when specific use case demands. - Configuration: - Use a reference to Azure Key Vault for retrieving secrets (e.g., database connection strings) in the Environment Variables section. - Use the App Configuration Service for non-secret configurations. - Enable Managed Identity. - Networking & Security: - Use VNET Integration. - Enable Access Restrictions to limit inbound traffic. - Monitoring: - Enable Diagnostic Settings and Application Insights.
Azure Databricks	Name: <i>ppg-adb-ppuk-lab-uks-<environment>-001</i> - Instances:1

Data Toolkit	Deployment Best Practice
	• Region: UK South • Environments: PPUK requires Azure Databricks deployed across the following environments; • Dev • Cluster Configuration: ○ Job Clusters: For scheduled workloads – D4a v4. ○ Interactive Clusters: For ad-hoc queries – D4a v4. ○ Enable the auto-termination parameter to stop clusters after X minutes of idle time. ○ SQL Compute: 2XS • Autoscaling: ○ Configure a minimum of 2 and a maximum of X nodes, depending on workload requirements. • Networking & Security: ○ Deploy with VNET Injection. ○ Use Azure Private Link to connect to storage. • Monitoring: ○ Enable Diagnostic Settings and Application Insights. • Best Practices: ○ Databricks Premium plan shall be used for role-based access control & enhanced security. ○ Deploy Azure Databricks in custom Vnet. ○ Azure AD be used for authentication and built in Databricks RBAC roles for platform management and access. ○ RBAC shall be used with least privilege model. ○ Access tokens shall be stored in Azure Key Vault. ○ Automated deployment through IaC. ○ Cluster policies shall be implemented to limit the ability to create clusters based on a set of rules. ○ Diagnostic logging must be enabled and configured as per Calandra Landing zone standards. ○ Resource Tagging for costing and billing must be applied in advance as per Calandra Landing zone policies.
Azure Storage Accounts	• Name: ○ For ADLS Gen 2 - <i>ppgadlspukdpuks<environment>001</i> ○ For Azure Blob Storage - <i>ppgabsppukaaeuks<environment>001</i> • Instances: 2 Accounts (see above) • Region: UK South • Environments: PPUK requires storage accounts across the following environments; • Dev • Test • UAT

Data Toolkit	Deployment Best Practice
	• Pre-Production • Production • Replication Type: ◦ LRS for non-production. ◦ ZRS for production. • Performance Tier: ◦ Hot Tier for frequently accessed data. • Security & Compliance: ◦ Enable Private Endpoints for secure access. ◦ Enable Storage Firewall to allow only trusted networks. ◦ Disable storage account key access so that access is only through Managed Identity. ◦ Use Customer-Managed Keys (CMK) for encryption if required. • Monitoring: ◦ Enable Diagnostic Settings and Application insights. • Create following storage accounts for Prod and non-Prod environments: ◦ Create one Azure ADLS Gen2 storage account for the Azure Databricks LAB environment. ◦ Create Azure Blob storage accounts, separate for Prod vs Non-prod, for internal use by the AAE team as per design.
Azure Machine Learning (Azure ML)	• Name: <i>ppg-aml-ppuk-aae-uks-<environment>-001</i> • Instances: TBD – depends on workspace strategy (see below) • Region: UK South • Environments: PPUK requires Azure ML Workspaces deployed across the following environments; • Dev • Test • UAT • Pre-Production • Production • Workspace Setup: ◦ Resource Group: Create a dedicated resource group for Azure ML resources. ◦ Workspace: Create an Azure ML workspace within the resource group. Consider separate workspaces for development, testing, and production. • Compute Resources: ◦ Compute Instances: Configure compute instances for interactive development and experimentation. ◦ Compute Clusters: Set up compute clusters for training machine learning models at scale. • Data Storage:

Data Toolkit	Deployment Best Practice
	○ Azure Blob Storage: Use Azure Blob Storage to store datasets, model artifacts, and other related files. ○ Leverage the Azure ML Feature Store for storing, managing and reusing engineered across models and pipelines. ○ Azure SQL Managed Instance: Read-only connection via Azure Private Link. • Model Registry: ○ Azure ML Model Registry: Utilize the model registry to manage and version machine learning models. • Model Deployment: ○ Use Azure ML managed online endpoints for real-time inference and batch endpoints for asynchronous scoring tasks. ○ Real-time or batch inference should be selected on a use-case basis to ensure effective cost management and performance of the solution. ○ Use endpoint traffic splitting to roll out new and approved model versions gradually and reduce risk. ○ Configure autoscaling for inference endpoints based on request volume and latency SLAs. • Model Monitoring: ○ Use Azure ML's out of box <i>data drift monitors</i> to track changes in data distributions between training and inference datasets. ○ Track datasets, training scripts, environments, and registered models via <i>Azure ML lineage tracking</i>. ○ Configure alerts when drift exceeds threshold, which can trigger retraining workflows via CI/CD pipelines. • CI/CD for ML: ○ MLOps: Implement MLOps practices to automate the training, deployment, and monitoring of machine learning models. Integrate with Azure DevOps for CI/CD pipelines. ○ Use automated triggers in Azure DevOps to promote models between environments. • Security: ○ Managed Identities: Use managed identities to grant Azure ML access to other Azure resources securely. ○ Role-Based Access Control (RBAC): Utilize RBAC to control access to Azure ML resources. ○ Use Azure ML Private Link to ensure isolation of network traffic for both training and compute endpoints. ○ Azure private link also used to connect Azure ML workspaces to the SQL Server Managed Instance (data platform).

Data Toolkit	Deployment Best Practice
	○ Consider use of Customer Managed Keys (CMK) as part of Azure Key Vault for higher compliance as per individual asset requirements ○ Disable public network access for the resource by default. ○ If public endpoints are required, restrict access to known IP ranges via firewall rules.